Modelling and control of multibody mechanical systems

Model answers

Question 1

a) i) The position vector of m_1 is

$$\boldsymbol{r_1} = -\frac{l}{2}\boldsymbol{e_\theta} + l_1\boldsymbol{e_{r_1}}$$

and the position vector of m_2 is

$$\boldsymbol{r_2} = \frac{l}{2}\boldsymbol{e_\theta} + l_2\boldsymbol{e_{r_2}}$$

. [2 marks]

ii) By differentiating each of the position vectors we obtain respectively the velocity vector of m_1 as

$$\dot{\boldsymbol{r}}_{\boldsymbol{1}} = \frac{l}{2}\dot{\theta}\boldsymbol{e_r} + l_1\left(\dot{\theta}+\dot{\theta_1}\right)\boldsymbol{e_{\theta_1}} = \frac{l}{2}\dot{\theta}\cos\theta_1\boldsymbol{e_{r_1}} + \left(-\frac{l}{2}\dot{\theta}\sin\theta_1 + l_1\left(\dot{\theta}+\dot{\theta_1}\right)\right)\boldsymbol{e_{\theta_1}}$$

and the velocity vector of m_2 as

$$\dot{\boldsymbol{r}}_{\boldsymbol{2}} = -\frac{l}{2}\dot{\theta}\boldsymbol{e_r} + l_2\left(\dot{\theta}+\dot{\theta_2}\right)\boldsymbol{e_{\theta_2}} = -\frac{l}{2}\dot{\theta}\cos\theta_2\boldsymbol{e_{r_2}} + \left(\frac{l}{2}\dot{\theta}\sin\theta_2 + l_2\left(\dot{\theta}+\dot{\theta_2}\right)\right)\boldsymbol{e_{\theta_2}},$$

since $\boldsymbol{e_r} = \cos\theta_1\boldsymbol{e_{r_1}} - \sin\theta_1\boldsymbol{e_{\theta_1}} = \cos\theta_2\boldsymbol{e_{r_2}} - \sin\theta_2\boldsymbol{e_{\theta_2}}$. Note that the angular velocity of the pendulums with respect to the inertial reference frame is, respectively, $\dot{\theta}+\dot{\theta_1}$ and $\dot{\theta}+\dot{\theta_2}$, and therefore these terms appear in the derivatives of $\boldsymbol{e_{r_1}}$ and $\boldsymbol{e_{r_2}}$, when the position vectors are differentiated to obtain the velocity vectors.

[4 marks]

Answers

b) i) The total kinetic energy of the system is

$$T = \frac{1}{2}m_1\dot{\boldsymbol{r}}_{\boldsymbol{1}}\cdot\dot{\boldsymbol{r}}_{\boldsymbol{1}} + \frac{1}{2}m_2\dot{\boldsymbol{r}}_{\boldsymbol{2}}\cdot\dot{\boldsymbol{r}}_{\boldsymbol{2}}$$

$$= \frac{1}{2}m_1\left(\frac{l^2}{4}\dot{\theta}^2 - ll_1\dot{\theta}\left(\dot{\theta}+\dot{\theta_1}\right)\sin\theta_1 + l_1^2\left(\dot{\theta}+\dot{\theta_1}\right)^2\right)$$

$$+\frac{1}{2}m_2\left(\frac{l^2}{4}\dot{\theta}^2 + ll_2\dot{\theta}\left(\dot{\theta}+\dot{\theta_2}\right)\sin\theta_2 + l_2^2\left(\dot{\theta}+\dot{\theta_2}\right)^2\right)$$

[2 marks]

ii) The potential energy of the system, with zero potential energy at the level of the point O, is

$$V = -m_1g\left(\frac{l}{2}\sin\theta + l_1\cos(\theta+\theta_1)\right) - m_2g\left(-\frac{l}{2}\sin\theta + l_2\cos(\theta+\theta_2)\right)$$

[3 marks]

iii) The Lagrangian function of the system is

$$L = T - V$$

$$= \frac{1}{2}m_1\left(\frac{l^2}{4}\dot{\theta}^2 - ll_1\dot{\theta}\left(\dot{\theta}+\dot{\theta_1}\right)\sin\theta_1 + l_1^2\left(\dot{\theta}+\dot{\theta_1}\right)^2\right)$$

$$+\frac{1}{2}m_2\left(\frac{l^2}{4}\dot{\theta}^2 + ll_2\dot{\theta}\left(\dot{\theta}+\dot{\theta_2}\right)\sin\theta_2 + l_2^2\left(\dot{\theta}+\dot{\theta_2}\right)^2\right)$$

$$+m_1g\left(\frac{l}{2}\sin\theta + l_1\cos(\theta+\theta_1)\right) + m_2g\left(-\frac{l}{2}\sin\theta + l_2\cos(\theta+\theta_2)\right)$$

[1 mark]

c) i) The Lagrangian equation with respect to the generalised coordinate θ_1 is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial\dot{\theta_1}}\right) - \frac{\partial L}{\partial\theta_1} = 0$$

which yields the first equation of motion

$$-\frac{l}{2}\ddot{\theta}\sin\theta_1 + \frac{l}{2}\dot{\theta}^2\cos\theta_1 + l_1\left(\ddot{\theta}+\ddot{\theta_1}\right) + g\sin\left(\theta+\theta_1\right) = 0$$

[3 marks]

Answers

ii) The Lagrangian equation with respect to the generalised coordinate θ_2 is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial\dot{\theta_2}}\right) - \frac{\partial L}{\partial\theta_2} = 0$$

which yields the second equation of motion

$$\frac{l}{2}\ddot{\theta}\sin\theta_2 - \frac{l}{2}\dot{\theta}^2\cos\theta_2 + l_2\left(\ddot{\theta}+\ddot{\theta_2}\right) + g\sin\left(\theta+\theta_2\right) = 0$$

[3 marks]

d) The Lagrangian equation with respect to the generalised coordinate θ is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial\dot{\theta}}\right) - \frac{\partial L}{\partial\theta} = 0$$

[2 marks]

Question 2

a) The position vector of the centre of mass of the drone is

$$\boldsymbol{r} = x\boldsymbol{i} + a\boldsymbol{e_r}$$

[3 marks]

b) The velocity vector is found by differentiating the position vector. The velocity vector of the centre of mass of the drone is

$$\dot{\boldsymbol{r}} = \dot{x}\boldsymbol{i} + a\dot{\theta}\boldsymbol{e_\theta} = \left(\dot{x} + a\dot{\theta}\cos\theta\right)\boldsymbol{i} - a\dot{\theta}\sin\theta\boldsymbol{k}$$

[3 marks]

c) The total kinetic energy of the drone is:

$$T = \frac{1}{2}m\dot{\boldsymbol{r}}\cdot\dot{\boldsymbol{r}} + \frac{1}{2}I\dot{\theta}^2$$

$$= \frac{1}{2}m\left(\dot{x}^2 + 2a\dot{x}\dot{\theta}\cos\theta + a^2\dot{\theta}^2\right) + \frac{1}{2}I\dot{\theta}^2$$

[3 marks]

d) The potential energy of the drone is

$$V = -mga\cos\theta$$

with the rail taken as the zero potential energy level.
Hence the Lagrangian is $L = T - V$, which gives

$$L = \frac{1}{2}m\left(\dot{x}^2 + 2a\dot{x}\dot{\theta}\cos\theta + a^2\dot{\theta}^2\right) + \frac{1}{2}I\dot{\theta}^2 + mga\cos\theta$$

Answers

[3 marks]

e) The Lagrangian equation with respect to the generalised coordinate θ is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial\dot{\theta}}\right) - \frac{\partial L}{\partial\theta} = T_{th},$$

which gives

$$ma\ddot{x}\cos\theta + \left(ma^2 + I\right)\ddot{\theta} + mga\sin\theta = T_{th}.$$

The Lagrangian equation with respect to the generalised coordinate x is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial\dot{x}}\right) - \frac{\partial L}{\partial x} = -F_{th}\sin\theta,$$

yielding

$$m\ddot{x} + ma\ddot{\theta}\cos\theta - ma\dot{\theta}^2\sin\theta = -F_{th}\sin\theta.$$

[8 marks]

Question 3

a) The generalised coordinates are θ and x. [2 marks]

b) The position vector of the centre of mass of the drone is

$$\boldsymbol{r} = x\boldsymbol{i} + a\boldsymbol{e_r}.$$

By differentiating twice the position vector, the acceleration vector of the centre of mass of the drone is given by

$$\ddot{\boldsymbol{r}} = \ddot{x}\boldsymbol{i} + a\ddot{\theta}\boldsymbol{e_\theta} - a\dot{\theta}^2\boldsymbol{e_r} = \left(\ddot{x} + a\ddot{\theta}\cos\theta - a\dot{\theta}^2\sin\theta\right)\boldsymbol{i} - \left(a\ddot{\theta}\sin\theta + a\dot{\theta}^2\cos\theta\right)\boldsymbol{k}$$

[3 marks]

c) The total force on the drone in vector form is

$$\boldsymbol{F_{total}} = F_v\boldsymbol{k} + mg\boldsymbol{k} - F_{th}\boldsymbol{e_r} = -F_{th}\sin\theta\boldsymbol{i} + (F_v + mg - F_{th}\cos\theta)\,\boldsymbol{k}$$

[3 marks]

d) By considering the translational motion of the centre of mass of the drone one has

$$-F_{th}\sin\theta\boldsymbol{i} + (F_v + mg - F_{th}\cos\theta)\,\boldsymbol{k} = m\left(\left(\ddot{x} + a\ddot{\theta}\cos\theta - a\dot{\theta}^2\sin\theta\right)\boldsymbol{i} - \left(a\ddot{\theta}\sin\theta + a\dot{\theta}^2\cos\theta\right)\boldsymbol{k}\right) \tag{1}$$

therefore by collecting the $\boldsymbol{k}$ terms

$$F_v = -m\left(a\ddot{\theta}\sin\theta + a\dot{\theta}^2\cos\theta + g\right) + F_{th}\cos\theta \tag{2}$$

Answers

[5 marks]

e) By collecting the $\boldsymbol{i}$ terms in equation (1), the first equation of motion is

$$-F_{th}\sin\theta = m\left(\ddot{x} + a\ddot{\theta}\cos\theta - a\dot{\theta}^2\sin\theta\right). \tag{3}$$

Considering the motion about the centre of mass of the drone according to $\frac{d\boldsymbol{H}}{dt} = \boldsymbol{N}$, where

$$H = I\dot{\theta},$$

and

$$N = a\sin\theta F_v + T_{th}$$

gives

$$I\ddot{\theta} = a\sin\theta F_v + T_{th}.$$

By substitution, using equations (2) and (3), the second equation of motion is

$$ma\ddot{x}\cos\theta + \left(ma^2 + I\right)\ddot{\theta} + mga\sin\theta = T_{th}.$$

[7 marks]

Question 4

a) 3 degrees of freedom, generalised coordinates are θ, θ_1 and θ_2. [2 marks]

b) By simple geometric considerations:

$$\begin{aligned} \boldsymbol{e_r} &= \cos\theta_1 \boldsymbol{e_{r_1}} - \sin\theta_1 \boldsymbol{e_{\theta_1}} \\ \boldsymbol{e_\theta} &= \sin\theta_1 \boldsymbol{e_{r_1}} + \cos\theta_1 \boldsymbol{e_{\theta_1}} \\ \boldsymbol{e_r} &= \cos\theta_2 \boldsymbol{e_{r_2}} - \sin\theta_2 \boldsymbol{e_{\theta_2}} \\ \boldsymbol{e_\theta} &= \sin\theta_2 \boldsymbol{e_{r_2}} + \cos\theta_2 \boldsymbol{e_{\theta_2}} \end{aligned} \tag{4}$$

[2 marks]

c) By differentiating the velocity vector of mass m_1 we obtain

$$\ddot{\boldsymbol{r}}_{\mathbf{1}} = \frac{l}{2}\ddot{\theta}\boldsymbol{e_r} + \frac{l}{2}\dot{\theta}^2\boldsymbol{e_\theta} + l_1\left(\ddot{\theta}+\ddot{\theta_1}\right)\boldsymbol{e_{\theta_1}} - l_1\left(\dot{\theta}+\dot{\theta_1}\right)^2\boldsymbol{e_{r_1}},$$

and by using equations (4)

$$\ddot{\boldsymbol{r}}_{\mathbf{1}} = \left(\frac{l}{2}\ddot{\theta}\cos\theta_1 + \frac{l}{2}\dot{\theta}^2\sin\theta_1 - l_1\left(\dot{\theta}+\dot{\theta_1}\right)^2\right)\boldsymbol{e_{r_1}} + \left(-\frac{l}{2}\ddot{\theta}\sin\theta_1 + \frac{l}{2}\dot{\theta}^2\cos\theta_1 + l_1\left(\ddot{\theta}+\ddot{\theta_1}\right)\right)\boldsymbol{e_{\theta_1}}$$

[3 marks]

Answers

d) By considering the motion of the mass m_1 one has

$$\left(m_1 g\cos(\theta+\theta_1) - F_{r_1}\right)\boldsymbol{e_{r_1}} - m_1 g\sin(\theta+\theta_1)\,\boldsymbol{e_{\theta_1}} = m_1\ddot{\boldsymbol{r}}_{\mathbf{1}}, \tag{5}$$

where F_{r_1} is the force of constraint in the pendulum rod holding mass m_1.

i) By collecting the $\boldsymbol{e_{\theta_1}}$ terms in equation (5), the first equation of motion of the pendulum system can be derived as follows

$$-\frac{l}{2}\ddot{\theta}\sin\theta_1 + \frac{l}{2}\dot{\theta}^2\cos\theta_1 + l_1\left(\ddot{\theta}+\ddot{\theta_1}\right) + g\sin(\theta+\theta_1) = 0$$

[3 marks]

ii) By collecting the $\boldsymbol{e_{r_1}}$ terms in equation (5), the force of constraint in the pendulum rod holding mass m_1 is given by

$$F_{r_1} = m_1\left(g\cos(\theta+\theta_1) - \frac{l}{2}\ddot{\theta}\cos\theta_1 - \frac{l}{2}\dot{\theta}^2\sin\theta_1 + l_1\left(\dot{\theta}+\dot{\theta_1}\right)^2\right)$$

[3 marks]

e) By differentiating the velocity vector of mass m_2 we obtain

$$\ddot{\boldsymbol{r}}_{\mathbf{2}} = -\frac{l}{2}\ddot{\theta}\boldsymbol{e_r} - \frac{l}{2}\dot{\theta}^2\boldsymbol{e_\theta} + l_2\left(\ddot{\theta}+\ddot{\theta_2}\right)\boldsymbol{e_{\theta_2}} - l_2\left(\dot{\theta}+\dot{\theta_2}\right)^2\boldsymbol{e_{r_2}},$$

and by using equations (4)

$$\ddot{\boldsymbol{r}}_{\mathbf{2}} = \left(-\frac{l}{2}\ddot{\theta}\cos\theta_2 - \frac{l}{2}\dot{\theta}^2\sin\theta_2 - l_2\left(\dot{\theta}+\dot{\theta_2}\right)^2\right)\boldsymbol{e_{r_2}} + \left(\frac{l}{2}\ddot{\theta}\sin\theta_2 - \frac{l}{2}\dot{\theta}^2\cos\theta_2 + l_2\left(\ddot{\theta}+\ddot{\theta_2}\right)\right)\boldsymbol{e_{\theta_2}}$$

[3 marks]

f) By considering the motion of the mass m_2 one has

$$(m_2 g \cos(\theta + \theta_2) - F_{r_2})\, \boldsymbol{e_{r_2}} - m_2 g \sin(\theta + \theta_2)\, \boldsymbol{e_{\theta_2}} = m_2 \ddot{\boldsymbol{r}}_{\boldsymbol{2}}, \tag{6}$$

where F_{r_2} is the force of constraint in the pendulum rod holding mass m_2.

i) By collecting the $\boldsymbol{e_{\theta_2}}$ terms in equation (6), the second equation of motion of the pendulum system can be derived as follows

$$\frac{l}{2}\ddot{\theta}\sin\theta_2 - \frac{l}{2}\dot{\theta}^2\cos\theta_2 + l_2\left(\ddot{\theta} + \ddot{\theta}_2\right) + g\sin(\theta + \theta_2) = 0$$

[2 marks]

ii) By collecting the $\boldsymbol{e_{r_2}}$ terms in equation (6), the force of constraint in the pendulum rod holding mass m_2 is given by

$$F_{r_2} = m_2\left(g\cos(\theta + \theta_2) + \frac{l}{2}\ddot{\theta}\cos\theta_2 + \frac{l}{2}\dot{\theta}^2\sin\theta_2 + l_2\left(\dot{\theta} + \dot{\theta}_2\right)^2\right)$$

[2 marks]

Answers